



**MISPA nano** *Plus*

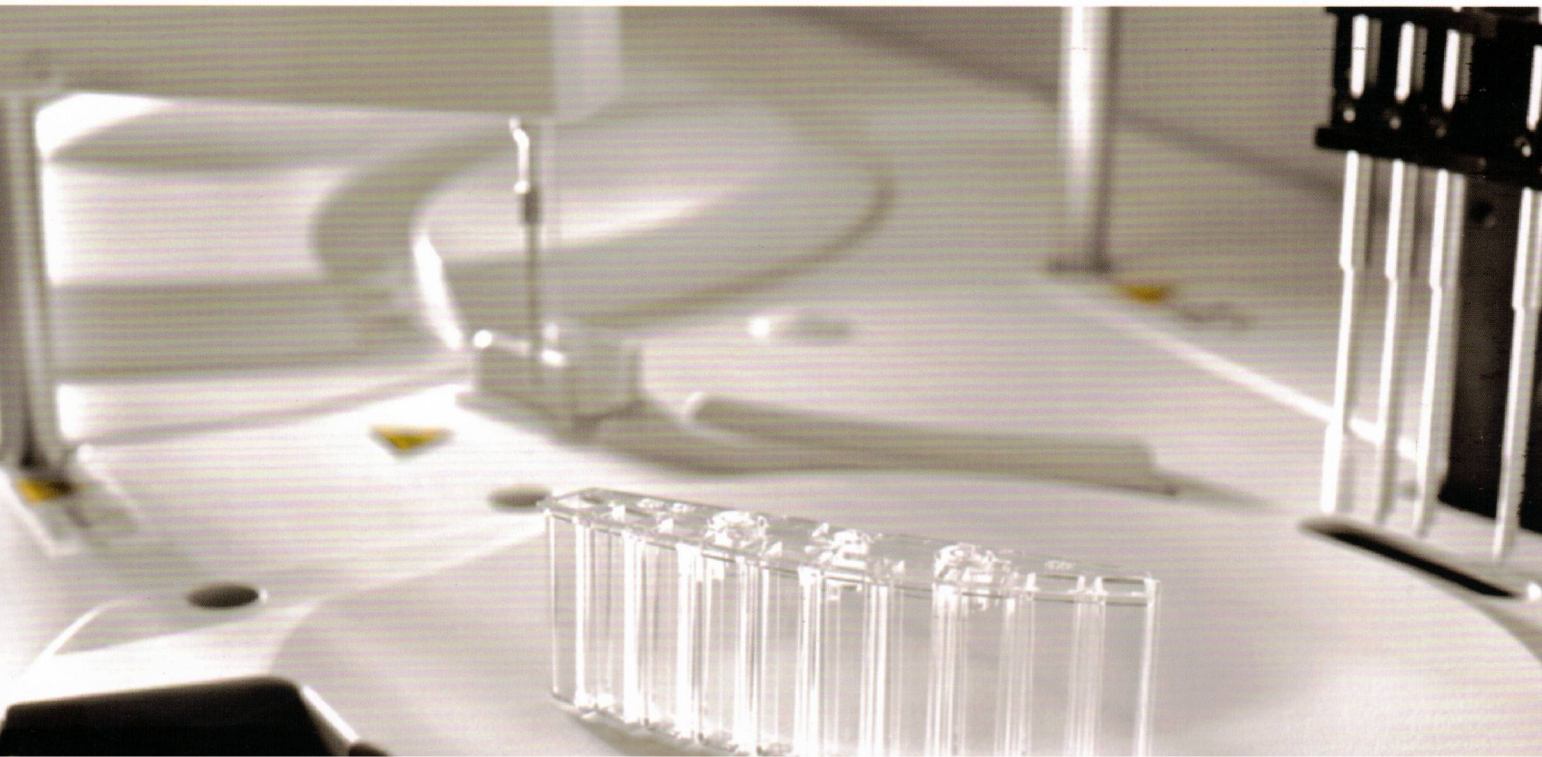
## Improved Speed and Performance

The most advanced and uniquely designed benchtop equipment for today's entry level to medium volume lab. Up to 360 tests per hour with all biochemistry and immunochemistry parameters. The reagent tray provides 24 hour non stop onboard refrigeration at 2 to 12°C. The sample positions can accommodate primary sample tubes, aliquot tubes and micro cups.



## New Generation Photometer

All the 80 reusable cuvettes are automatically cleaned by the 4 stage auto wash station. The optical system utilizes Grating photometry with twelve wavelengths from 340 to 800 nm to accommodate most of the clinical chemistry testing requirements with a minimum reaction volume of 100µL and ensures highest accuracy.



# MISPA nano<sup>Plus</sup>

Fully Automatic Clinical Chemistry Analyser

Optimized V



Waterfall probe  
cleaning



Intelligent probe with optional  
clot detection



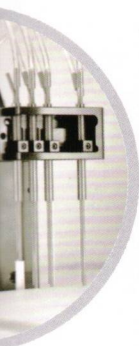
Double switch  
for power saving



ENGINEERING A

# MISPA nano<sup>Plus</sup>

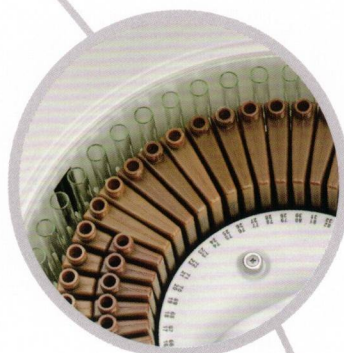
Fully Automatic Clinical Chemistry Analyser



Sample Loading Station



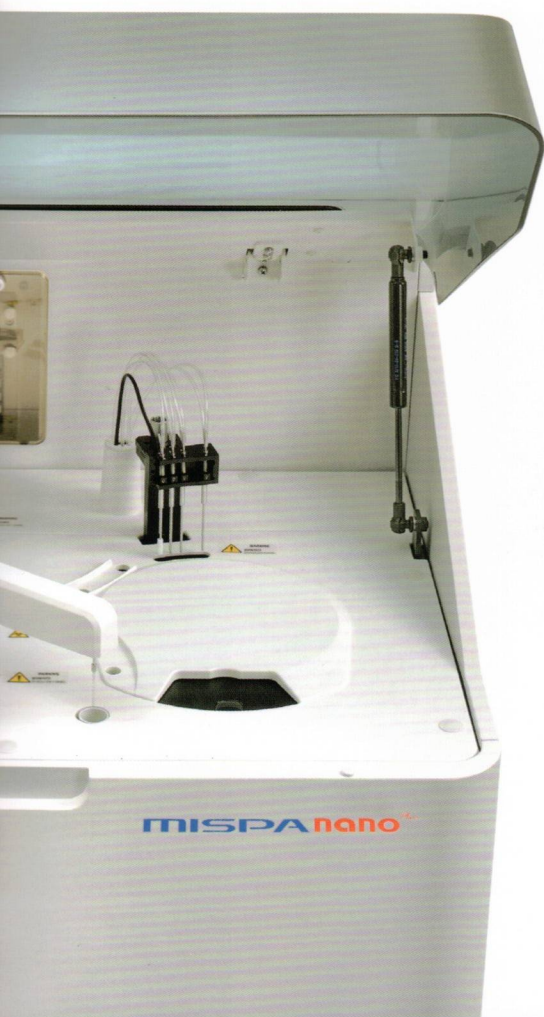
Independent mixing bar



Built-in barcode reader



Intuitive software with more functionalities



NEW MOMENTUM

# MISPA nano<sup>Plus</sup>

Fully Automatic Clinical Chemistry Analyser

## Considerate Design

Smaller Footprint | Robust Functionality  
User-Friendly designs | Long Uptime.

Up to 100 sample positions  
Up to 100 reagent positions  
(50 fixed + 50 interchangeable)



## Basic Renal Profile

- Urea U V
- Jaffe Creatinine
- Uric Acid

## Advanced Renal Profile

- Enzymatic Creatinine
- Urea U V
- Uric Acid

## Extended Renal Profile

- Calcium
- Magnesium
- MALB
- C3, C4
- Cystatin C

## Basic Diabetic Profile

- Glucose
- HbA1c Direct

## Advanced Diabetic Profile

- Glucose Hexokinase

## Cardiac Profile

- CK-NAC
- hsCRP

## Basic Liver Function Tests

- Alkaline Phosphatase
- Bilirubin Direct
- Bilirubin Total
- Total Protein
- Albumin
- SGOT
- SGPT

## Extended LFT

- Gamma GT

## Basic Lipid Profile

- Cholesterol
- Triglycerides
- HDL-C Direct
- LDL-C Direct

## Extended Lipid Profile

- Lp(a)
- Apo A1, Apo B

## RA Profile

- Rheumatoid Factor
- C-Reactive Protein

## Immunoglobulins

- IgA
- IgE
- IgG
- IgM

## Pancreatic Tests

- $\alpha$ - Amylase

## Other Parameters

- Phosphorus
- LDH-P
- ASO
- Ceruloplasmin
- Ferritin
- ...



# Technical Specifications

|                                                                     |                                                                                                                                                                                    |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Function</b>                                              |                                                                                                                                                                                    |
| Automatic, Discrete, Random Access, Bench-top, STAT sample priority |                                                                                                                                                                                    |
| Throughput:                                                         | Up to 360 photometric tests per hour                                                                                                                                               |
| Measuring principles:                                               | Absorbance photometry, turbidimetry                                                                                                                                                |
| Methodology:                                                        | End-point, Fixed-time, Kinetic, Single/Double reagent chemistries, Mono-chromatic / bi-chromatic                                                                                   |
| <b>Reagent/Sample Handling</b>                                      |                                                                                                                                                                                    |
| Reagent/Sample tray:                                                | 50 to 100 positions for reagents and 50 to 100 positions for samples in 24-hour refrigerated compartment (2~12℃)                                                                   |
| Reagent volume:                                                     | R1: 100~200μL, step by 0.5μL; R2: 10~200μL, step by 0.5μL                                                                                                                          |
| Sample volume:                                                      | 2~35μL, step by 0.1μL                                                                                                                                                              |
| Reagent/Sample probe:                                               | Liquid level detection, vertical collision protection, inventory checking, reagent pre-warming, optional clot detection                                                            |
| Probe cleaning:                                                     | Automatic washing for interior and exterior; Carry over < 0.05%                                                                                                                    |
| Automatic sample dilution:                                          | Pre-dilution and post-dilution                                                                                                                                                     |
| Mixing Unit:                                                        | Independent mixing bar                                                                                                                                                             |
| Capable to communicate with LIS in bi-directional mode              |                                                                                                                                                                                    |
| <b>Reaction System</b>                                              |                                                                                                                                                                                    |
| Reaction tray:                                                      | 80 reusable cuvettes                                                                                                                                                               |
| Reaction volume                                                     | 100~360μL                                                                                                                                                                          |
| Reaction temperature:                                               | 37℃±0.1℃                                                                                                                                                                           |
| Cuvette Washing:                                                    | Washing station with pre-warmed detergent and de-ionized water                                                                                                                     |
| <b>Optical System</b>                                               |                                                                                                                                                                                    |
| Light Source:                                                       | Halogen-tungsten lamp                                                                                                                                                              |
| Wavelength:                                                         | 12 wavelengths, 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm, 800nm (Grating Photometry)                                                            |
| Absorption range:                                                   | 0~3.5Abs, resolution 0.0001Abs                                                                                                                                                     |
| <b>Control and Calibration</b>                                      |                                                                                                                                                                                    |
| Calibration modes:                                                  | K factor, Linear (two points and multipoints), Logit-Log 4P, Logit-Log 5P, Spline, Exponential, Polynomial, Parabola, Logit-Log3P, Broken line, One key calibrator import function |
| Control Rules:                                                      | Westgard multi-rule, Levey-Jennings, Cumulative sum check, Twin plot                                                                                                               |
| <b>Operation Unit</b>                                               |                                                                                                                                                                                    |
| Operation system:                                                   | Windows 10                                                                                                                                                                         |
| Interface:                                                          | RS-232                                                                                                                                                                             |
| <b>Working Conditions</b>                                           |                                                                                                                                                                                    |
| Power Supply:                                                       | 200~240V, 50/60Hz, 1300VA or 100~130V, 60Hz, 1300VA                                                                                                                                |
| Dimension:                                                          | 860 mm (length) × 660 mm (depth) × 550 mm (height)                                                                                                                                 |
| Weight:                                                             | 115 kg                                                                                                                                                                             |
| Water Consumption:                                                  | 6.5 L/hour                                                                                                                                                                         |

EN ISO 13485:2016



Toll Free No: 1800 425 7151